

C4 photosynthesis dependence with temperature and hydraulic

Chilperic Armel Foko Kuate, Jérémie Muller-Prokob, Antonio Rigueiro, Martin Lercher*

* Correspondence: Martin Lercher (martin.lercher@hhu.de)

Institute for Computational Cell Biology, Heinrich Heine University, Düsseldorf, Germany

Cluster of Excellence on Plant Sciences (CEPLAS), Heinrich Heine University, Düsseldorf,
Germany

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Abstract

Introduction

Aims of the method

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Discussion and conclusions

Perspectives

- Highlight importance of the field:
- Summary of the current thinking:
- Comment on future directions:

17 **Abbreviations**

18 F

19 **Author contributions**

20 Conceptualization C.A.F.K.,A.R.,M.L.; Methodology C.A.F.K., A.R., M.L; Investigation C.A.F.K., Writ-
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26 **Competing Interests**

27 The Authors declare that there are no competing interests associated with the manuscript.

28 Appendix

29 List of symbols

30	A	Activator
31	I	Inhibitor
32	K_m	Michaelis-Menten constant
33	K_a	Activation constant
34	K_i	Inhibition constant
35	k_{cat}	Turnover number
36	V_{max}	Maximum velocity of the reaction
37	$K_{M_{Acet}}$	Michaelis-Menten constant for acetyl-CoA
38	$K_{M_{Mal}}$	Michaelis-Menten constant for malonyl-CoA
39	$K_{M_{NADPH}}$	Michaelis-Menten constant for NADPH
40	$K_{I_{Acet}}$	Inhibitor constant for acetyl-CoA
41	$K_{I_{Mal}}$	Inhibitor constant for malonyl-CoA
42	$Acet$	Acetyl-CoA
43	Mal	Malonyl-CoA

44 1 Model description

45 We assume that the plant is a continuum of three major compartments, namely the mesophyll cell,
46 the bundle sheath cell, and the soil. the compartments exchange heat to make the temperature of the
47 relevant compartment suitable to perform the main tasks of the compartment. For instance, making the
48 bundle sheat temperature suitable for rubisco maximum assimilation rate as rubisco kinetic parameter
49 depend on the leaf temperature.

50 1.1 Assumptions of the model

Parameter Name	Formula	Value
Leaf temperature	T_{bs}	Variable (K)
Specific heat capacity	C_p	4.18 J/(g·K)
Water density	ρ	1000 kg/m ³
Leaf volume	V	Varies (m ³)
Time	t	Time (s)
Albedo	α	0.3 (dimensionless)
Solar radiation flux	ψ	1000 W/m ²
Stefan-Boltzmann constant	σ	5.67×10^{-8} W/(m ² ·K ⁴)
Emissivity	ε	0.95 (dimensionless)
Optimal control variable for water flux	F	Variable (W/K)
Soil temperature	T_{soil}	290 K
Convective heat transfer coefficient	h	10 W/(m ² ·K)
Air temperature	T_{air}	298 K
Latent heat of vaporization	λ	2.45 MJ/kg
Transpiration rate	E	Variable (kg/m ² ·s)

Table 1: Parameters of the differential equation for leaf temperature

51 The differential equations governing the mesophyll temperature are giving by:

$$C_p \rho V \frac{dT_{mes}}{dt} = (1 - \alpha) \psi - \sigma \varepsilon T_{mes}^4 - UA(T_{mes} - T_{bs}) - h(T_{mes} - T_{air}) - \lambda E. \quad (1.1)$$

The bundle sheath temperature is described by the following ODE

$$C_p \rho V \frac{dT_{bs}}{dt} = UA(T_{mes} - T_{bs}) - FC_p(T_{bs} - T_{soil}) \quad (1.2)$$

- The soil temperature (T_{soil}), solar irradiance (ψ), the transpiration rate (E) are considered to be constant
- the flux of water acting for cooling (F) is represented as a control variable
- The leaf temperature (T_l) is the variable solution to the steady state equation of energy balance of the leaf:

At the steady state, the T_{mes} , T_{bs} are solutions of the system of algebraic equations:

$$\begin{cases} (1 - \alpha)\psi - \sigma \varepsilon T_{mes}^4 - UA(T_{mes} - T_{bs}) - h(T_{mes} - T_{air}) - \lambda E = 0 \\ UA(T_{mes} - T_{bs}) - FC_p(T_{bs} - T_{soil}) = 0 \end{cases}$$

2 Objective

- Given the irradiance and the soil temperature Find the optimal control F (Flux) that make T_{bs} suitable for rubisco assimilation rate (Rub)

3 The steady-state energy balance in the leaf

At the steady state, the energy balance of the leaf is governed by the exchange of heat between the inner cells (mesophyll and bundle sheath)

$$\alpha\phi - UA(T_{mes} - T_{bs}) - \sigma(T_{mes}^4 - T_{sky}^4) - h(T_{mes} - T_{air}) - \lambda E = 0 \quad (e1)$$

We can rearrange equation (e1) to express it as a polynomial in T_{mes} . This yields:

$$-\sigma T_{mes}^4 - (UA + h)T_{mes} + (\alpha\phi + UAT_{bs} + \sigma T_{sky}^4 + hT_{air} - \lambda E) = 0 \quad (3.1)$$

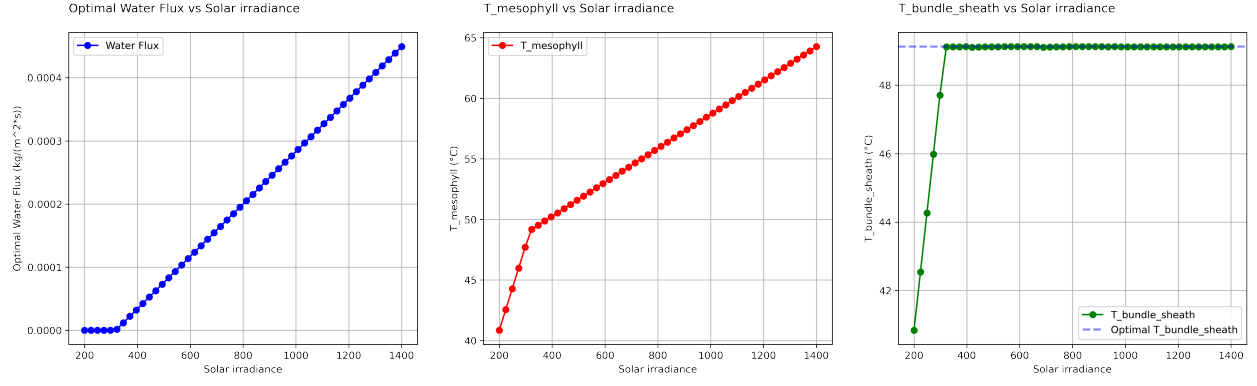


Figure 1: Solar irradiance effect on mesophyll and bundle-sheath temperatures, as well as the flux of water necessary to regulate the bundle sheath temperature optimal for rubisco assimilation

Here, the coefficients of the polynomial are given by:

$$a = -\sigma \quad b = -(UA + h) \quad c = \alpha\phi + UA T_{bs} + \sigma T_{sky}^4 + h T_{air} - \lambda E$$

Next, consider the second expression:

$$UA(T_{mes} - T_{bs}) = CF(T_{mes} - T_{soil}) \quad (e2)$$

Solving for T_{bs} , we obtain:

$$T_{bs} = T_{mes} - \frac{CF(T_{mes} - T_{soil})}{UA} \quad (3.2)$$

Given that the parameters $\alpha, \phi, UA, \sigma, h, \lambda, E, T_{sky}, T_{air}, T_{soil}, C$ are known, except for the function F , knowing T_{mes} allows T_{bs} to be entirely determined by optimizing F . Conversely, knowing T_{bs} allows for the full determination of T_{mes} .

Therefore, the interdependence of T_{mes} and T_{bs} is established through the optimization of the function F , given the known parameters.

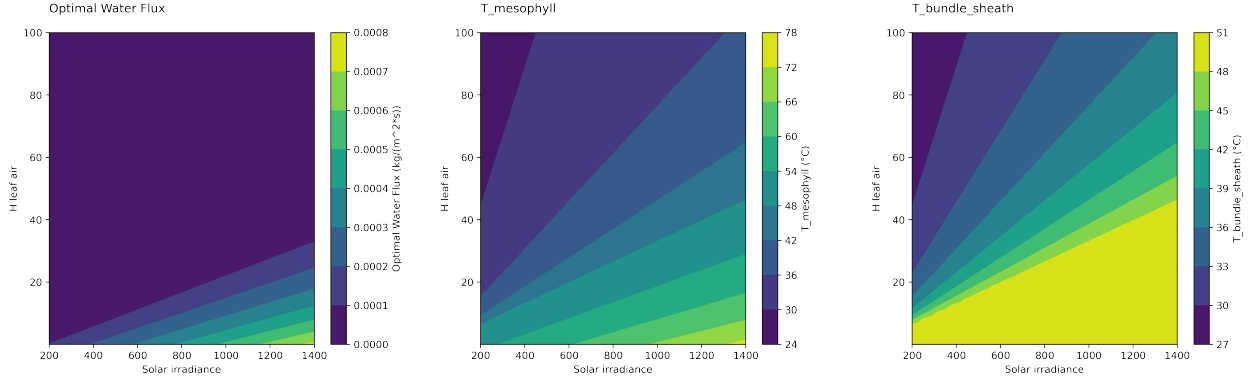


Figure 2: Combination of the effect of solar irradiance and the heat convective transfer coefficient on mesophyll and bundle-sheath temperatures, as well as the flux of water necessary to regulate the bundle sheath temperature optimal for rubisco assimilation

4 Analytical approach for the steady-state model

4.1 The Energy Balance of the Mesophyll and Bundle-Sheath Cells

Governing Equations

The governing equation for the temperature of the mesophyll cells is given by:

$$C\rho V \frac{dT_{mes}}{dt} = (1 - \alpha)\phi - UA(T_{mes} - T_{bs}) - \sigma(T_{mes}^4 - T_{sky}^4) - h_{mes}(T_{mes} - T_{air}) - \lambda_{mes}E, \quad (4.1)$$

and for the bundle-sheath cells is given by:

$$C\rho V \frac{dT_{bs}}{dt} = UA(T_{mes} - T_{bs}) - CF(T_{bs} - T_{soil}), \quad (4.2)$$

where

- C : Specific heat capacity of the mesophyll and bundle-sheath cells (assumed to be water for now) ($J \cdot kg^{-1} \cdot K^{-1}$)
- ρ : Density of the mesophyll and bundle-sheath cells (assumed to be water for now) ($kg \cdot m^{-3}$)
- V : Volume of the mesophyll cells (m^3)
- T_{mes} : Temperature of the mesophyll cells (K)
- t : Time (s)

- 82 • α : Albedo of the mesophyll cells (dimensionless)
- 83 • β : Albedo of the bundle sheath cells (dimensionless)
- 84 • σ : Stefan-Boltzmann constant ($\text{W} \cdot \text{m}^{-2} \cdot \text{K}^{-4}$)
- 85 • ϕ : Solar radiation absorbed by the mesophyll cells ($\text{W} \cdot \text{m}^{-2}$)
- 86 • UA : Heat transfer coefficient between the mesophyll cells and the bundle-sheath cells per area
87 ($\text{W} \cdot \text{m}^{-2} \cdot \text{K}^{-1}$)
- 88 • T_{bs} : Temperature of the bundle-sheath cells (K)
- 89 • h_{mes} : Heat transfer coefficient between the mesophyll cells and the air ($\text{W} \cdot \text{m}^{-2} \cdot \text{K}^{-1}$)
- 90 • T_{air} : Temperature of the air (K)
- 91 • T_{sky} : Temperature of the sky (K)
- 92 • T_{soil} : Temperature of the soil (K)
- 93 • $F = v \cdot \rho$: Flux of water multiplied by water density (exchanging heat between the bundle-sheath
94 cells and the soil via water) ($\text{kg} \cdot \text{s}^{-1}$)
- 95 • λ_{mes} : Latent heat of vaporization of water in the mesophyll cells ($\text{J} \cdot \text{kg}^{-1}$)
- 96 • E : Rate of evaporation of water from the mesophyll cells ($\text{kg} \cdot \text{s}^{-1}$)

97 Steady-State Equations

98 At steady state, the temperature of the mesophyll and bundle-sheath cells are constant, leading to the
99 following equations:

For mesophyll cells:

$$(1 - \alpha)\phi - UA(T_{mes} - T_{bs}) - \sigma(T_{mes}^4 - T_{sky}^4) - h_{mes}(T_{mes} - T_{air}) - \lambda_{mes}E = 0 \quad (4.3)$$

For bundle-sheath cells:

$$UA(T_{mes} - T_{bs}) - CF(T_{bs} - T_{soil}) = 0 \quad (4.4)$$

Expressing T_{bs} in terms of the water flux F :

$$T_{bs} = \frac{UA T_{mes} + CF T_{soil}}{UA + CF} \quad (4.5)$$

By substituting equation 4.5 into 4.3, we get:

$$(1-\alpha)\phi - UA \left(T_{mes} - \frac{UA T_{mes} + CF T_{soil}}{UA + CF} \right) - \sigma(T_{mes}^4 - T_{sky}^4) - h_{mes}(T_{mes} - T_{air}) - \lambda_{mes}E = 0, \quad (4.6)$$

which can be rewritten:

$$(1-\alpha)\phi - \left(\frac{UACF}{UA + CF} + h_{mes} \right) T_{mes} + \frac{UACFT_{soil}}{UA + CF} - \sigma T_{mes}^4 + \sigma T_{sky}^4 + h_{mes}T_{air} - \lambda_{mes}E = 0 \quad (4.7)$$

0 which can be rewritten as 4th degree polynomial equation in T_{mes} :

$$\sigma T_{mes}^4 + \left(\frac{UACF}{UA + CF} + h_{mes} \right) T_{mes} - (1-\alpha)\phi - \frac{UACFT_{soil}}{UA + CF} - \sigma T_{sky}^4 - h_{mes}T_{air} + \lambda_{mes}E = 0 \quad (4.8)$$

Polynomial Representation and Analysis

Consider the polynomial:

$$\sigma T_{mes}^4 + \left(\frac{UACF}{UA + CF} + h_{mes} \right) T_{mes} - (1-\alpha)\phi - \frac{UACFT_{soil}}{UA + CF} - \sigma T_{sky}^4 - h_{mes}T_{air} + \lambda_{mes}E = 0 \quad (4.9)$$

We rewrite it as follows:

$$aT_{mes}^4 + bT_{mes} + c = 0, \quad (4.10)$$

where the coefficients are:

$$\begin{aligned} a &= \sigma, \\ b &= \left(\frac{UACF}{UA + CF} + h_{mes} \right), \\ c &= -(1-\alpha)\phi - \frac{UACFT_{soil}}{UA + CF} - \sigma T_{sky}^4 - h_{mes}T_{air} + \lambda_{mes}E. \end{aligned}$$

To analyze the possible number of positive roots of 4.10, one needs Descartes' Rule of Signs.

4.2 Descartes' Rule of Signs

Descartes' Rule of Signs provides a way to determine the number of positive and negative real roots of a polynomial by examining the number of sign changes in the sequence of its coefficients.

Positive Roots

For positive roots, Descartes' Rule of Signs states that the number of positive real roots of a polynomial is equal to the number of sign changes between consecutive non-zero coefficients or fewer than it by an even number.

Negative Roots

For negative roots, Descartes' Rule of Signs states that the number of negative real roots of a polynomial is determined by applying the rule to $P(-x)$, the polynomial formed by substituting $-x$ for x in the original polynomial. The number of sign changes in the sequence of coefficients of $P(-x)$ gives the number of negative real roots, or fewer than it by an even number.

Application to the Polynomial

Positive Roots

For our polynomial, we examine the signs of the coefficients:

- $a = \sigma > 0$
- $b = \left(\frac{UACF}{UA + CF} + h_{mes} \right) > 0$
- c can be either positive or negative

Case: $c > 0$

If $c > 0$, the polynomial $aT_{mes}^4 + bT_{mes} + c$ has no sign changes because all coefficients a , b , and c are positive. According to Descartes' Rule of Signs, this implies there are zero positive roots.

125 **Case:** $c < 0$

126 If $c < 0$, the polynomial $aT_{mes}^4 + bT_{mes} + c$ has one sign change between bT_{mes} and the constant term
127 c . This suggests the possibility of one positive root.

128 Negative Roots

129 To determine the number of negative roots, we consider the polynomial $P(-T_{mes})$:

$$\sigma(-T_{mes})^4 + \left(\frac{UACF}{UA + CF} + h_{mes} \right) (-T_{mes}) - (1 - \alpha)\phi - \frac{UACFT_{soil}}{UA + CF} - \sigma T_{sky}^4 - h_{mes}T_{air} + \lambda_{mes}E = 0 \quad (4.11)$$

130 Simplifying, we get:

$$\sigma T_{mes}^4 - \left(\frac{UACF}{UA + CF} + h_{mes} \right) T_{mes} - (1 - \alpha)\phi - \frac{UACFT_{soil}}{UA + CF} - \sigma T_{sky}^4 - h_{mes}T_{air} + \lambda_{mes}E = 0 \quad (4.12)$$

131 We examine the signs of the coefficients:

- 132 • $a = \sigma > 0$
- 133 • $b = - \left(\frac{UACF}{UA + CF} + h_{mes} \right) < 0$
- 134 • c can be either positive or negative

135 **Case:** $c > 0$

136 If $c > 0$, the polynomial $aT_{mes}^4 - bT_{mes} + c$ has one sign change between a and $-b$. This suggests the
137 possibility of one negative root.

138 **Case:** $c < 0$

139 If $c < 0$, the polynomial $aT_{mes}^4 - bT_{mes} + c$ has two sign changes, suggesting the possibility of two or
140 zero negative roots.

141 Finding a Condition on F for $c < 0$

Given:

$$c = -(1 - \alpha)\phi - \frac{UACFT_{soil}}{UA + CF} - \sigma T_{sky}^4 - h_{mes}T_{air} + \lambda_{mes}E \quad (4.13)$$

To ensure $c < 0$:

$$\lambda_{mes}E < (1 - \alpha)\phi + \frac{UACFT_{soil}}{UA + CF} + \sigma T_{sky}^4 + h_{mes}T_{air} \quad (4.14)$$

142 Isolating F , 4.14 can be rewritten as follow:

$$F > \frac{\left(\lambda_{mes}E - (1 - \alpha)\phi - \sigma T_{sky}^4 - h_{mes}T_{air}\right) UA}{UACT_{soil} - \left(\lambda_{mes}E - (1 - \alpha)\phi - \sigma T_{sky}^4 - h_{mes}T_{air}\right)} \quad (4.15)$$

143 Summary Table

Case	c Sign	Number of Positive Solutions
$c > 0$	Positive	0
$c < 0$	Negative	<u>1</u>

144 5 Update from the discussion of October 9, 2024

—> According to the literature [[Jeremie update](#)], there are more chloroplasts in the bundle sheath than in the mesophyll, which suggests that ETC may also take place in the bundle sheath. In this case, one must consider the irradiance absorbed by the bundle sheath compartment. So, one should modify the previous version of the model to include a term $((1 - \beta)\phi)$ encoding for the irradiance absorbed by the bundle sheath. So we modify (4.2) and (4.4)

$$C\rho V \frac{dT_{bs}}{dt} = (1 - \beta)\phi - UA(T_{mes} - T_{bs}) - CF(T_{bs} - T_{soil}), \quad (5.1)$$

for the dynamic version and

$$(1 - \beta)\phi + UA(T_{mes} - T_{bs}) - CF(T_{bs} - T_{soil}) = 0 \quad (5.2)$$

145 for the steady-state equation. In the above equations, β is the albedo value for the bundle sheath.

At steady-state, T_{bs} can be written as:

$$T_{bs} = \frac{(1 - \beta)\phi + UA T_{mes} + CF T_{soil}}{UA + CF}. \quad (5.3)$$

Replacing T_{bs} by its expression in (4.3), and after some algebra, one obtain:

$$\sigma T_{mes}^4 - T_{mes} \left(h_{mes} + \frac{UACF}{UA + CF} \right) - \left((1 - \alpha)\phi - \frac{UA((1 - \beta)\phi + CF T_{soil})}{UA + CF} + h_{mes} T_{air} + \lambda_{mes} E \right) = 0, \quad (5.4)$$

which can be written as

$$aT_{mes}^4 + bT_{mes} + c = 0, \quad (5.5)$$

with

$$a = \sigma$$

,

$$b = - \left(h_{mes} + \frac{UACF}{UA + CF} \right)$$

$$c = - \left((1 - \alpha)\phi - \frac{UA((1 - \beta)\phi + CF T_{soil})}{UA + CF} + h_{mes} T_{air} + \lambda_{mes} E \right)$$

Steady-State Model of Heat Management in the C4 Plant

6 Model Description and Components

In this section, we update the steady-state model of heat management in the C4 plant. This version of the model incorporates four compartments instead of the two considered previously. In addition to the mesophyll and bundle-sheath temperatures (T_{mes} , T_{bs}), we now include equations for the temperature balance in the soil. This temperature balance is represented by T_{surf} and T_{root} , which denote the soil temperatures at the surface and the root level, respectively. This two-layer soil representation follows the approach outlined in prior work [1].

6.1 Description of Soil Temperature

In our approach, the soil temperature distribution is divided into three compartments or layers based on depth (source and methodology for depth selection to be included). The temperature at the soil surface, T_{surf} , is defined from 0 to h_1 m depth. The temperature at the root level, T_{root} , spans from h_1 to h_2 m. Beyond h_2 m, the final compartment, T_{∞} , represents the deep soil temperature. While the first two compartments have temperatures that may vary during the diurnal cycle, the temperature in the last compartment is nearly constant and can therefore be treated as a fixed parameter.

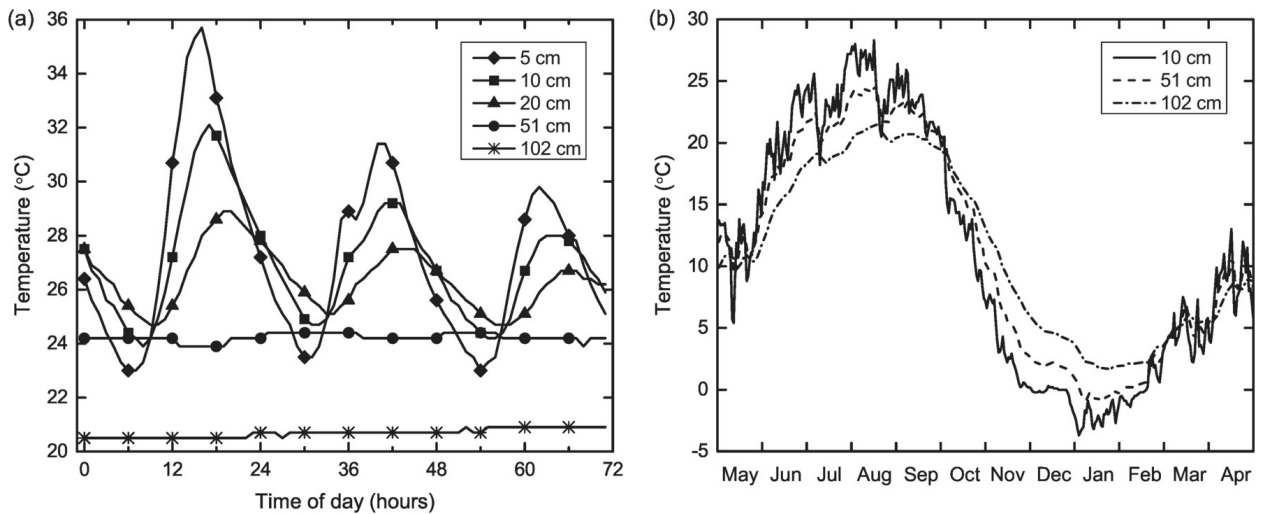


Figure 3: Distribution of soil temperature at different depths during diurnal and seasonal cycles.

T_{surf} receives direct solar irradiance, which is moderated by the albedo γ , accounting for cloud cover
 and plant canopy effects. This attenuation reduces the radiation reaching the surface. Part of the
 absorbed radiation is lost through radiative and convective cooling. Additionally, T_{surf} exchanges heat
 conductively with T_{bs} and T_{root} . Similarly, T_{root} interacts thermally with T_{∞} while being influenced by
 heat conduction from T_{surf} .

6.2 Dynamic model

The heat management system in the plant is described by a set of four ordinary differential equations
 (ODEs), each representing the thermal balance of a specific compartment:

- **Mesophyll:**

$$C_w v_{leaf} \frac{dT_{mes}}{dt} = (1 - \alpha)\phi - UA(T_{mes} - T_{bs}) - \sigma\epsilon_{leaf}T_{mes}^4 - h(T_{mes} - T_{air}) - \lambda E_{leaf}, \quad (6.1)$$

- **Bundle-sheath:**

$$C_w v_{leaf} \frac{dT_{bs}}{dt} = (1 - \beta)\phi + UA(T_{mes} - T_{bs}) - C_w F(T_{bs} - T_{root}), \quad (6.2)$$

- **Soil surface:**

$$C_s v_{soil} \frac{dT_{surf}}{dt} = (1 - \gamma)\phi - h(T_{surf} - T_{air}) - K_{surf}(T_{surf} - T_{root}) - \sigma\epsilon_{soil}T_{surf} - \lambda E_{soil}, \quad (6.3)$$

- **For Soil-roots**

$$C_s v_{soil} \frac{dT_{surf}}{dt} = K_{surf}(T_{surf} - T_{root}) + C_w F(T_{bs} - T_{root}) - K_{root}(T_{root} - T_{inf}). \quad (6.4)$$

7 Remarks

- Shall we consider soil evaporation as a certain quantity of heat is lost through soil water evaporation

Symbol	Description	Units
T_{mes}	Temperature of mesophyll cells	(K)
T_{bs}	Temperature of bundle-sheath	(K)
T_{surf}	Temperature of soil surface	(K)
T_{root}	Temperature of soil-root	(K)
$F = v \cdot \rho$	Flux of water multiplied by density	$(\text{kg} \cdot \text{s}^{-1})$
E	Rate of evaporation of water	$(\text{kg} \cdot \text{s}^{-1})$

Table 2: **Summary of variables, and decision variables:** the variables in teal are the key variables describing the state of the system. The variables in magenta are decisions variables of the optimization problem

172 7.1 Note on the radiative cooling: linear radiative cooling

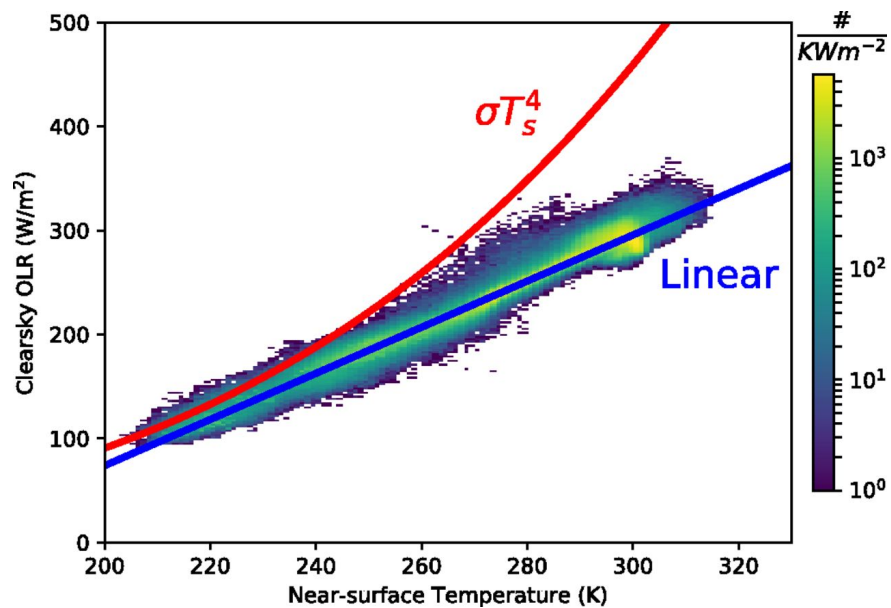


Figure 4: Global radiative cooling of the earth seems to be linear as shown by Koll and Cronin

Symbol	Description	Units
C_w	Specific heat capacity of water (for Mes and BS)	$(\text{J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1})$
C_s	Specific heat capacity of the soil	$(\text{J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1})$
ρ	water volumetric mass	$(\text{J} \cdot \text{kg} \cdot \text{m}^{-3})$
α	Albedo of mesophyll cells (dimensionless)	$(-)$
β	Albedo of bundle-sheath cells (dimensionless)	$(-)$
γ	Albedo of soil (dimensionless)	$(-)$
σ	Stefan-Boltzmann constant	$(\text{W} \cdot \text{m}^{-2} \cdot \text{K}^{-4})$
ϕ	Solar radiation	$(\text{W} \cdot \text{m}^{-2})$
UA	Conductive heat transfer coefficient (Mes→ BS)(per area)	$(\text{W} \cdot \text{m}^{-2} \cdot \text{K}^{-1})$
h	convective heat transfer coefficient	$(\text{W} \cdot \text{m}^{-2} \cdot \text{K}^{-1})$
K_{surf}	conductive heat transfer coefficient (soil surf → soil-root)	$(\text{W} \cdot \text{m}^{-2} \cdot \text{K}^{-1})$
K_{root}	conductive heat transfer coefficient (soil-root → soil deeper layers)	$(\text{W} \cdot \text{m}^{-2} \cdot \text{K}^{-1})$
T_{air}	Temperature of air	(K)
T_{sky}	Temperature of sky	(K)
T_{soil}	Temperature of soil	(K)
T_{∞}	Temperature of soil deeper layers	(K)
λ	Latent heat of vaporization of water	$(\text{J} \cdot \text{kg}^{-1})$
E_{soil}	Evaporation rate for the soil	$(\text{Kg} \cdot \text{s}^{-1})$

Table 3: Summary of parameters of the model, descriptions, and units

8 steady state model

From the dynamic model represented by (6.1), (6.2), (6.3), and (6.4) one can derive the steady-state model by setting their left-hand sides to 0. Hence, one obtains:

- Mesophyll:

$$(1 - \alpha)\phi - UA(T_{mes} - T_{bs}) - \sigma\epsilon_{leaf}T_{mes}^4 - h(T_{mes} - T_{air}) - \lambda E_{leaf} = 0, \quad (8.1)$$

- Bundle-sheath:

$$(1 - \beta)\phi + UA(T_{mes} - T_{bs}) - C_w F(T_{bs} - T_{root}) = 0, \quad (8.2)$$

- **Soil surface:**

$$(1 - \gamma)\phi - h(T_{surf} - T_{air}) - K_{surf}(T_{surf} - T_{root}) - \sigma\epsilon_{soil}T_{surf} - \lambda E_{soil} = 0, \quad (8.3)$$

- **For Soil-roots**

$$K_{surf}(T_{surf} - T_{root}) + C_w F(T_{bs} - T_{root}) - K_{root}(T_{root} - T_{inf}) = 0. \quad (8.4)$$

176 If we consider the linear radiative heat transfer, the steady-state balance equation for mesophyll and soil
177 surface temperatures becomes:

$$(1 - \alpha)\phi - UA(T_{mes} - T_{bs}) - r(T_{mes} - T_{sky}) - h(T_{mes} - T_{air}) - \lambda E_{leaf} = 0, \quad (8.5)$$

$$(1 - \gamma)\phi - h(T_{surf} - T_{air}) - K_{surf}(T_{surf} - T_{root}) - r(T_{surf} - T_{sky}) - \lambda E_{soil} = 0, \quad (8.6)$$

178 where r is a value to be fitted from data.

179 8.1 Remarks

180 Currently we are not considering the effect of soil transpiration that influences soil temperature and
181 therefore the whole temperature balance of the plant.

182 9 Results

183 The first step in deriving results with the constructed steady-state model is to parameterize the model.

184 We choose to collect the parameter values or/and range for Sudan as described by the table below:

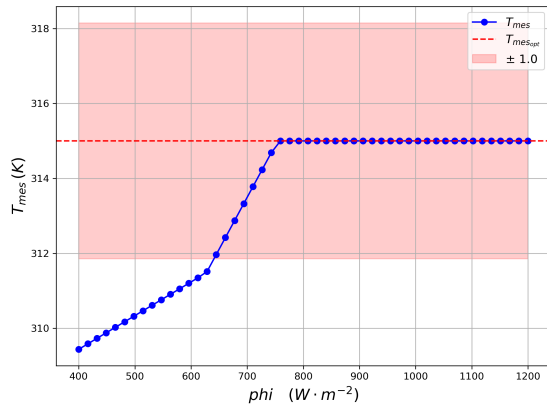


Figure 5: Caption for the first image

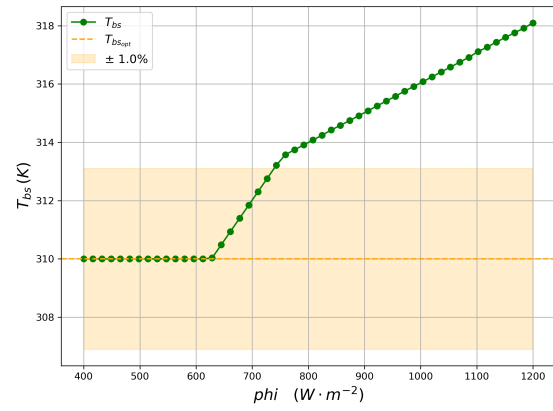


Figure 6: Caption for the second image

Symbol	Units	Value for Sudan	Sources
C_w	$(\text{J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1})$	4,186	[3]
C_s	$(\text{J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1})$	Approximately 800 to 1,200 (varies with soil composition)	[4, 5]
ρ	$(\text{kg} \cdot \text{m}^{-3})$	1,000	[3]
α	(—)	Typically ranges from 0.2 to 0.3	[6]
β	(—)	Typically ranges from 0.2 to 0.3	[6]
γ	(—)	Approximately 0.25 to 0.35 (varies with soil type and moisture content)	[4, 7]
σ	$(\text{W} \cdot \text{m}^{-2} \cdot \text{K}^{-4})$	5.67×10^{-8}	[3]
ϕ	$(\text{W} \cdot \text{m}^{-2})$	Varies; peak values can reach up to 1,000 during midday in Sudan	[8]
UA	$(\text{W} \cdot \text{m}^{-2} \cdot \text{K}^{-1})$	Specific to plant anatomy; typically ranges from 0.5 to 1.5	[6]
h	$(\text{W} \cdot \text{m}^{-2} \cdot \text{K}^{-1})$	Depends on wind speed; typically between 5 and 25	[9, 10]
K_{surf}	$(\text{W} \cdot \text{m}^{-2} \cdot \text{K}^{-1})$	Varies with soil type and moisture; generally between 0.5 and 2.5	[4, 7]
K_{root}	$(\text{W} \cdot \text{m}^{-2} \cdot \text{K}^{-1})$	Varies with soil type and moisture; generally between 0.5 and 2.5	[4, 7]
T_{air}	(K)	Varies; average daily temperatures in Sudan range from 293 to 313 (20°C to 40°C)	[11]
T_{sky}	(K)	Typically lower than air temperature; can be estimated based on atmospheric conditions	[12]
T_{soil}	(K)	Varies with depth and time of day; surface temperatures can exceed 323	[4, 7]

9.1 Notes on Convective Cooling

Convective cooling describes the influence of the surrounding temperature on the material of interest, in our case the leaf and the soil surface. Convective cooling is expressed as the product of the temperature difference between the material of interest and the surroundings (air temperature) and the convective heat transfer coefficient (h). The coefficient h depends on the wind speed (v_{wind}). Several empirical equations have been used in the literature to describe the relationship between h and v_{wind} . Below are a few equations that have been recorded

1. Monteith and Unsworth (2008):

$$h_c = 6.62 \cdot v_{wind}^{0.67}$$

Source: Monteith and Unsworth [10].

Comments:

- Suitable for general agricultural and natural canopies.
- Accounts for moderately turbulent forced convection.
- Commonly used in SPAC models.

2. Thom (1972):

$$h_c = 5.8 + 7.4 \cdot v_{wind}$$

Source: Thom [14].

Comments:

- Widely applied for canopy systems with moderate wind speeds.
- Linear relationship makes it computationally convenient.
- Includes both natural and forced convection contributions.

3. Grassland Canopy (Short Vegetation):

$$h_c = 10 + 6 \cdot v_{wind}$$

Source: Grace [9].

Comments:

- Empirical fit for short vegetation with minimal turbulence.

- Wind effects dominate heat transfer at higher speeds.

4. Forest Canopy (Tall Vegetation):

$$h_c = 15 + 4 \cdot v_{wind}$$

Source: Monteith and Unsworth [10].

Comments:

- Represents canopies with significant roughness and turbulence.
- h_c increases slower with v_{wind} compared to shorter vegetation.

General Comments:

- **Wind Speed Dependency:** All equations highlight that h_c increases with wind speed, reflecting the dominance of forced convection in higher winds.
- **Vegetation Type:** Shorter canopies exhibit steeper increases in h_c with u , while taller, rougher canopies (e.g., forests) increase more gradually due to turbulence.
- **Application in SPAC Models:**
 - Choose the equation based on canopy height and type.
 - Ensure u reflects above-canopy wind speeds, as in-canopy dynamics differ.
 - Monteith and Unsworth (2008) is the most versatile for general purposes.

10 Updated Steady-State Model of C4 Photosynthesis

10.1 Introduction

C4 photosynthesis enhances photosynthetic efficiency in high-light, high-temperature, and low-CO₂ environments. The updated steady-state model by von Caemmerer [15] refines the representation of biochemical processes and intercellular transport mechanisms in C4 plants, providing an accurate framework for understanding and predicting photosynthetic performance.

10.2 Mathematical Framework of the Model

The model focuses on the biochemical and biophysical processes occurring in the mesophyll and bundle-sheath cells. It integrates enzyme kinetics, metabolite transport, and light reactions into a coherent mathematical framework.

10.2.1 Net Rate of CO₂ Fixation

The net rate of CO₂ fixation for C4 photosynthesis can be described by two equations. The first equation describes Rubisco carboxylation in the bundle sheath:

$$A = V_c - 0.5V_o - R_d \quad (10.1)$$

where V_c and V_o are the rates of Rubisco carboxylation and oxygenation, respectively, and R_d is the rate of mitochondrial respiration not associated with photorespiration.

10.2.2 Mitochondrial Respiration

Mitochondrial respiration can occur in both the mesophyll and bundle sheath. Since Rubisco can refix q'CO₂ released in the bundle sheath, R_d is given by:

$$R_d = R_{d,m} + R_{d,bs} \quad (10.2)$$

where $R_{d,m}$ and $R_{d,bs}$ are the respiration rates in the mesophyll and bundle sheath, respectively.

242 10.2.3 Mesophyll Reactions

243 CO₂ assimilation rate can also be written in terms of the mesophyll reactions:

$$A = V_p - R_m - L \quad (10.3)$$

244 where V_p is the rate of PEP carboxylation, R_m is the mitochondrial respiration occurring in the mesophyll,
245 and L is the rate of CO₂ leakage from the bundle sheath to the mesophyll.

246 10.2.4 CO₂ Leakage

247 The leak rate, L , is given by:

$$L = g_{bs}(C_s - C_m) \quad (10.4)$$

248 where g_{bs} is the conductance to CO₂ leakage, and C_s and C_m are the bundle sheath and mesophyll CO₂
249 partial pressures, respectively.

250 10.2.5 Energy Costs and Leakiness

251 The C₄ cycle consumes additional energy during PEP regeneration, and leakage of CO₂ from the bundle
252 sheath represents an energy cost. Leakiness (ϕ) is defined as:

$$\phi = \frac{L}{V_p} \quad (10.5)$$

253 Overcycling, another related term, defines the leak rate as a fraction of the CO₂ assimilation rate:

$$\phi_{\text{over}} = \frac{L}{A} \quad (10.6)$$

254 10.2.6 Enzyme-Limited Rate Equations

255 Under high irradiance, Rubisco carboxylation can be described by its RuBP-saturated rate:

$$V_c = V_{c\max} \frac{C_s}{C_s + K_c(1 + O_s/K_o)} \quad (10.7)$$

256 The ratio of oxygenation to carboxylation is given by:

$$\frac{V_o}{V_c} = \frac{O_s}{\Gamma^*} \quad (10.8)$$

257 where Γ^* is the CO_2 compensation point in the absence of mitochondrial respiration:

$$\Gamma^* = \frac{0.5O_s}{S_{c/o}} \quad (10.9)$$

258 where $S_{c/o}$ is Rubisco specificity.

259 10.2.7 Overall CO_2 Assimilation Rate

260 Combining the expressions, the Rubisco-limited rate of CO_2 assimilation is:

$$A = \frac{V_{c\max}(C_s - \Gamma^*)}{C_s + K_c(1 + O_s/K_o)} \quad (10.10)$$

261 Using C_m and O_m , the mesophyll CO_2 and O_2 partial pressures, we can derive an expression for C_s :

$$C_s = C_m - \frac{A}{g_{bs}} \quad (10.11)$$

262 10.2.8 PEP Carboxylation

263 When CO_2 is limiting, the rate of PEP carboxylation is:

$$V_p = \frac{V_{p\max}C_m}{C_m + K_p} \quad (10.12)$$

264 where $V_{p\max}$ is the maximum PEP carboxylation rate, and K_p is the Michaelis constant for CO_2 .

10.2.9 Light- and Electron Transport-Limited Rate Equations

The energy requirements for the regeneration of RuBP in the bundle sheath are the same as in a C3 leaf. However, there is an additional cost of 2 mol ATP for the regeneration of 1 mol PEP from pyruvate in the mesophyll:

$$\gamma = \frac{3}{2}V_c + 2V_p \quad (10.13)$$

where γ is the energy requirement due to photorespiration.

The rate of NADP consumption is given by the requirement of the C3 cycle:

$$\gamma_{\text{NADPH}} = \frac{V_c + V_o}{2} \quad (10.14)$$

10.2.10 Electron Transport and ATP Production

NADPH and ATP are produced by chloroplast electron transport. The rate of electron transport, J , is related to the absorbed irradiance, I_2 :

$$J = \frac{I_2 + J_{\max} - \sqrt{(I_2 + J_{\max})^2 - 4\theta I_2 J_{\max}}}{2\theta} \quad (10.15)$$

where $J_{\max}$ is the maximum electron transport rate and θ is an empirical curvature factor.

The rate of ATP production is given by:

$$v_{\text{ATP}} = \frac{J(1 - f_{\text{cyc}})}{h} \quad (10.16)$$

where h is the number of protons required per ATP generated, and f_{cyc} is the fraction of cyclic electron transport.

10.3 Summary of Variables and Parameters

10.4 Model Validation and Applications

The updated model has been validated using experimental data from various C4 species, showing high accuracy in predicting photosynthetic rates under different environmental conditions. Applications of this model include:

- Predicting the impact of climate change on C4 plant productivity.
- Guiding genetic engineering efforts to optimize photosynthetic efficiency in crop plants.
- Understanding the biochemical and physiological adaptations of C4 plants to diverse environmental stresses.

10.5 Conclusion

The updated steady-state model of C4 photosynthesis by Susanne von Caemmerer represents a significant enhancement in our comprehension of this complex pathway. By integrating new empirical findings and refining the kinetic parameters of crucial enzymes and transporters, the model provides a detailed and accurate framework for studying C4 photosynthesis. This model not only advances theoretical knowledge but also has practical implications for improving crop efficiency and resilience in the face of global climate change.

11 Temperature dependence of rubisco kinetic parameter

11.1 Turnover number

The standard way of modeling the dependence kinetic parameter with temperature is to use the Arrhenius equation or one of its variants. Galmes and co have done intensive work on this topic, see Galmés et al. [16], Hermida-Carrera et al. [17], and Perdomo et al. [18] group has an intensive work on the subject. The standard equation used is

$$k(T) = \exp \left(c - \frac{\Delta H_a}{RT} \right) \quad , \quad (11.1)$$

however, this equation failed to capture the enzyme deactivation for the high temperature and therefore can only represent the temperature dependence of the turnover number $k_{cat}(T)$ for temperature below the the threshold of deactivation. To circumvent this limitation, one can use the corrected version of the Arrhenius equation that includes the deactivation as proposed by [16],

$$k_{cat}(T) = \frac{\exp\left(c - \frac{\Delta H_a}{RT}\right)}{1 + \exp\left(\frac{\Delta ST - \Delta H_d}{RT}\right)} \tag{11.2}$$

where c is a constant, ΔH_a is the activation energy, ΔH_d is the deactivation energy, ΔS is the entropy, R is the gas constant, and T is the temperature in Kelvin.

Add the table here

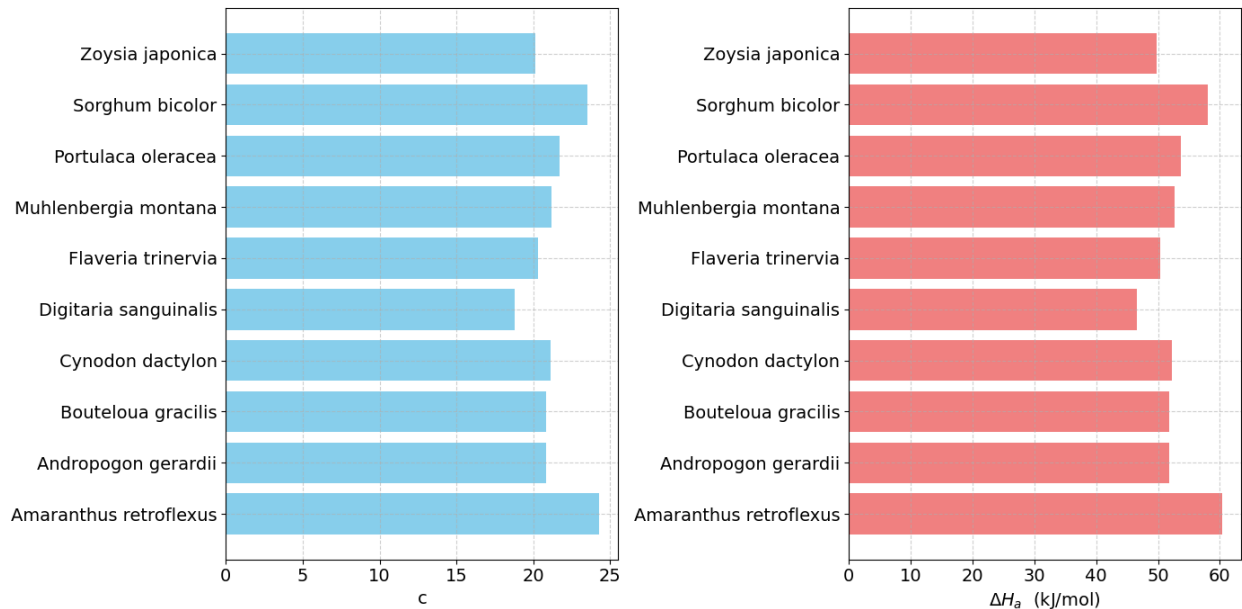


Figure 7: Caption

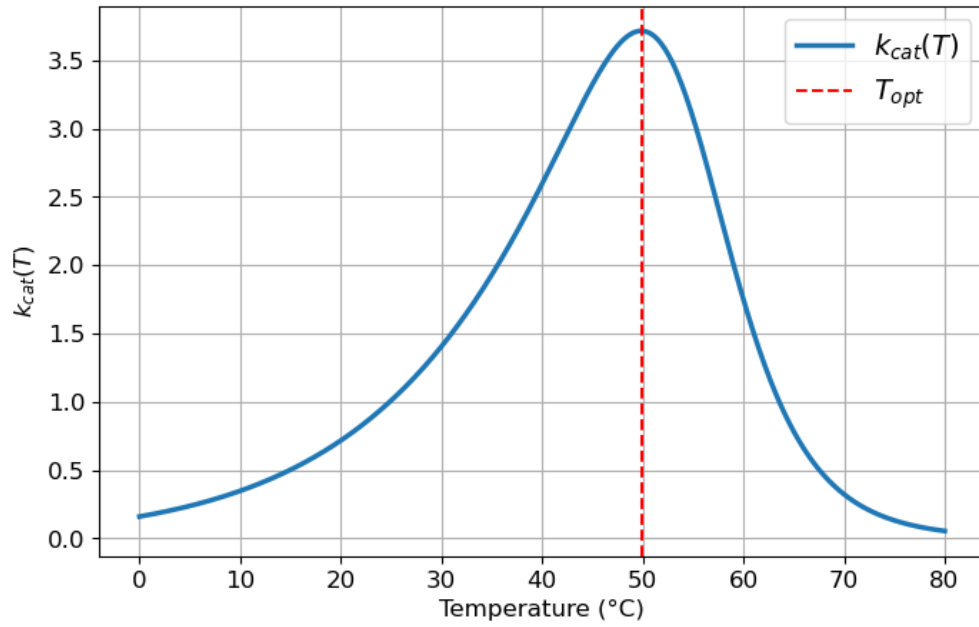


Figure 8: Caption

11.2 Michaelis constant

as mentioned above, the standard way of modeling the dependence of kinetic parameters on the temperature is to use the modified Arrhenius equation (see 11.1). A 2015 work by Perdomo et al., use a different version of Arrhenius like-equation:

$$P = P_{25} \exp \left(\frac{(T - 25)\Delta H_a}{298R(T + 273)} \right), \quad (11.3)$$

where the temperature here are in Celsius.

In the original paper they used E_a instead of ΔH_a .

Hermida-Carrera et al. use the same formulation as Galmés et al.. So the use of one other formulation will depends on the available data. One could thing of conciliating the two approaches by comparing the kinetic dependance with the temperature.

Boyd et al. [19] consider the following version of the Arrhenius equation to model the temperature response of kinetic parameters of the carbonic anhydrase and PEPC.

$$P(T) = P_{25} \cdot \exp \left(\frac{-E_a}{R \cdot T_k} \cdot \frac{298.15 - T_k}{298.15} \right). \quad (11.4)$$

For taking into account the effect of temperature on rubisco

$$P(T) = P_{25} \cdot \exp\left(\frac{E_a(T_k - 298.15)}{298.15 \cdot R \cdot T_k}\right) \cdot \frac{1 + \exp\left(\frac{298.15 \cdot \Delta S - H_d}{298.15 \cdot R}\right)}{1 + \exp\left(\frac{T_k \cdot \Delta S - H_d}{R \cdot T_k}\right)}, \quad (11.5)$$

- 305 • T_k : The temperature in Kelvin.
- 306 • $P(T_k)$: The parameter value at temperature T_k
- 307 • P_{25} : The parameter value at the reference temperature of 25°C (298.15 K)
- 308 • E_a : The activation energy
- 309 • R : The universal gas constant.
- 310 • $k(T_k)$: The rate constant at temperature T_k
- 311 • k_{25} : The rate constant at the reference temperature of 25°C (298.15 K)
- 312 • ΔS : The entropy change
- 313 • H_d : The enthalpy change (deactivation energy).

314 **Function Documentation:** `solve_quartic`

315 **Purpose**

316 The `solve_quartic` function symbolically solves a general quartic polynomial equation of the form:

$$Ax^4 + Bx^3 + Cx^2 + Dx + E = 0$$

317 Quartic equations (fourth-degree polynomials) can have up to four real or complex roots. This function
318 implements a symbolic solution using a method derived from the classical approach to solving quartics,
319 but simplified for practical computational use.

320 **Parameters**

- 321 • **A** (`sympy.Symbol`): The coefficient of the x^4 term in the quartic equation.
- 322 • **B** (`sympy.Symbol`): The coefficient of the x^3 term in the quartic equation.
- 323 • **C** (`sympy.Symbol`): The coefficient of the x^2 term in the quartic equation.
- 324 • **D** (`sympy.Symbol`): The coefficient of the x term in the quartic equation.
- 325 • **E** (`sympy.Symbol`): The constant term in the quartic equation.

326 All coefficients can be symbolic expressions or numeric values, depending on the problem at hand.

327 **Returns**

- 328 • **roots** (`List[sympy.Expr]`): A list of symbolic expressions representing the four possible roots of
329 the quartic equation. These roots can be real or complex, depending on the coefficients.

Methodology

1. Normalization

The function begins by dividing the entire quartic equation by A (assuming A is non-zero), thus simplifying the equation to:

$$x^4 + ax^3 + bx^2 + cx + d = 0$$

where $a = \frac{B}{A}$, $b = \frac{C}{A}$, $c = \frac{D}{A}$, and $d = \frac{E}{A}$.

2. Reduction to Depressed Quartic

The function does not explicitly reduce the equation to a depressed quartic (which eliminates the cubic term), but instead directly solves for roots using symbolic expressions derived from classical solutions.

3. Finding the Auxiliary Variable v_1

The roots of the quartic equation depend on the solution of an auxiliary quadratic equation in a variable v , which is solved first. The quadratic equation for v is given by:

$$v^2 + pv + q = 0$$

where p and q are expressions derived from the coefficients a , b , c , and d .

4. Calculation of u

Using the roots v_1 of the quadratic equation, an intermediate variable u is calculated. This involves a cube root of unity ζ , typically set to 1 for simplicity.

5. Determination of the Roots

The roots of the original quartic equation are determined using the calculated values of u and the coefficients of the normalized equation. The roots are given by the expressions:

$$x_{1,2} = -\frac{a}{4} \pm \sqrt{\frac{u}{2} + \text{Expression 1} \pm \text{Expression 2}}$$

$$x_{3,4} = -\frac{a}{4} \pm \sqrt{\frac{u}{2} - \text{Expression 3} \pm \text{Expression 4}}$$

Expression 1:

$$\text{Expression 1} = \frac{1}{4}(a^2 - 8b) - \frac{1}{4} \frac{a^3 - 4ab + 8c}{4\sqrt{u}}$$

348 This term comes from the discriminant and other algebraic manipulations that simplify the quartic into
349 a solvable form.

Expression 2:

$$\text{Expression 2} = \frac{-a^3 + 4ab - 8c}{4\sqrt{u}}$$

350 This expression results from completing the square and other algebraic steps during the derivation.

351 6. Return

352 The function returns a list of four roots, which are symbolic expressions that can be real or complex.

353 Example Usage

```
354 import sympy as sp
355
356 # Define the symbols for the coefficients
357 A, B, C, D, E = sp.symbols('A B C D E')
358
359 # Solve the quartic equation
360 roots = solve_quartic(A, B, C, D, E)
361
362 # Display the roots
363 for i, root in enumerate(roots, 1):
364     sp.pprint(f"Root {i}: {root}")
```

365 This function can be applied to specific quartic equations by substituting numeric values or expressions
366 for A , B , C , D , and E .

367 Notes

- 368 • The function uses `sympy`, a Python library for symbolic mathematics, allowing it to handle both
369 exact symbolic computations and numerical approximations.
- 370 • The roots may be complex depending on the nature of the coefficients, and further analysis might
371 be required to interpret the results in a specific context.

12 Notes from reading

- From von Caemmerer [15] at high temperature (approximately ≥ 22) the CO_2 assimilation rate is limited by the electron transport chain (see Fig. 6 in the paper).
- In C3 plants about 90% of the electron transport chain (ETC) is linear (Antonio)
- ETC can be divided into two: the linear ETC and the circular ETC.
- The linear ETC happens in the mesophyll cell while the circular ETC happens in the bundle sheath. The latter serves for the regeneration of Ribulose-1,5-bisphosphate

12.1 Note on Arrhenius' equation

On the Arrhenius' equation

$$k(T) = \exp\left(c - \frac{\Delta H_a}{RT}\right) \quad , \quad (12.1)$$

It is important to note that the temperature (T) is in Kelvin and the unit of gas constant R is $\text{Jmol}^{-1}\text{K}^{-1}$. Therefore to match the units, one needs to enter ΔH_a or ΔH_d in Jmol^{-1} . The value of the parameter obtained for a given temperature must be compared with actual data to validate the units.

12.2 Notes on C4 model and 4th degree polynomial

Bounding E : Since we cannot remove more energy than was absorbed,

$$\lambda_{mes}E \leq (1 - \alpha)\Phi \Rightarrow E \leq \frac{(1 - \alpha)\phi}{\lambda_{mes}E} \quad (12.2)$$

with

$$(1 - \alpha)\Phi \in \frac{[300, 1100]}{\lambda_{mes}} \Rightarrow E \leq 1.10^{-3} \quad (12.3)$$

On circular flow of water : The theory behind the project being to look for an evolutionary advantage to having Kranz anatomy in a plant, the main point of interest is the cooling capacity of the water flow from Xylem and Phloem. Thus, two questions :

- Is the heat exchange from the water flow, combined with other sources of heat exchanges in the leaf enough to achieve optimal temperatures for the relevant enzymes ?

Table 9-3. Representative Values for the Various Components of the Water Potential in the Soil–Plant–Atmosphere Continuum^a.

Location	Ψ or Ψ_{wp} (MPa)	P (MPa)	$-\Pi$ (MPa)	$\rho_w gh$ (MPa)	$\frac{RT}{V_w} \ln \frac{RH}{100}$ (MPa)
Soil 0.1 m below ground and 10 mm from root	-0.3	-0.2	-0.1	0.0	0.0
Soil adjacent to root	-0.5	-0.4	-0.1	0.0	0.0
Xylem of root near the ground surface	-0.6	-0.5	-0.1	0.0	0.0
Xylem in leaf at 10 m above the ground	-0.8	-0.8	-0.1	0.1	0.0
Vacuole of mesophyll cell in leaf at 10 m	-0.8	0.2	-1.1	0.1	0.0
Cell wall of mesophyll cell at 10 m	-0.8	-0.4	-0.5	0.1	
Air in cell wall pores at 10 m (water vapor assumed to be in equilibrium with water in cell wall)	-0.8			0.1	-0.9
Air just inside stomata at 95% relative humidity	-6.9			0.1	-7.0
Air just outside stomata at 60% relative humidity	-70.0			0.1	-70.1
Air just across boundary layer at 50% relative humidity	-95.1			0.1	-95.2

^a Ψ is equal to $P - \Pi + \rho_w gh$ in the liquid phases (Eq. 2.13a), and Ψ_{wp} is equal to $(RT/V_w) \ln (\% \text{ relative humidity}/100) + \rho_w gh$ in the gas phases (Eq. 2.24), all at 25 °C.

Figure 9: Representative Values for the Various Components of the Water Potential in the Soil Plant Atmosphere Continuum

- What drives the water flow to the leaf? → As we assume that transpiration is a less important factor than in C3 plants, the difference in water potential between low and high points in the plant should be lower as well.

12.3 Notes on water flow

Poiseuille's law for water flow in a tube

$$J_v = -\frac{r^2}{8\eta} \frac{\Delta P}{\Delta x} \quad (12.4)$$

The two main drivers of water flow in the Phloem (mainly top to bottom) are Osmotic pressure (due to the abundance of photosynthetic components in the phloem) and gravity (?)

Models have been developed to better understand how phloem flow works. The model from Thompson et al. [20] follows and tries to validate the Münch hypothesis of water flow in phloem. The Münch hypothesis proposes that water flow in the phloem is driven and regulated by the concentration gradient of solutes. According to this hypothesis, the plant secretes metabolites in the phloem (sources) to decrease osmotic pressure and consumes them from the phloem (sinks) to increase osmotic pressure. Thus creating a concentration gradient from the top of the plant to the bottom of the plant.

12.4 Heat exchange with radiation

We can bound the value of heat exchange :

$$\sigma(T_{mes}^4 - T_{mes}^4) \leq \sigma(\max(T_{mes}^4) - \min(T_{sky}^4)) = 463.53W \Rightarrow \sigma(T_{mes}^4 - T_{mes}^4) \leq 463.53W \quad (12.5)$$

With a maximal temperature for the Mesophyll of 322 °K and a minimal temperature for the sky of 223°K

12.5 Notes on 28.08 Meeting

$$q = \min(UA_{bs}(T_{bs} - T_{soil}), FC_p\rho(T_{bs} - T_{soil})). \quad (12.6)$$

Given that the first entry of the min function is not the limiting term, one can always bound F as follows

:

$$F \leq \frac{UA_{bs}}{C_p\rho} \quad (12.7)$$

1 Where C_p is the water capacity and ρ the water density.

12.5.1 Notes on optimal temperature

The optimal temperature of the Bundle Sheath cell and of the Mesophyll cell are set to the optimal temperature of the relevant enzymes that are active in each cell. For the Mesophyll cell, the selected enzyme is the Electron Transport Chain (ETC), with an optimal temperature of 298°K. For the Bundle Sheath cell, the selected enzyme is Rubisco, with an optimal temperature of 320°K. The optimal temperature of both cells raises a question : Knowing that the heat source is closer to the mesophyll cell and the cooling source is closer to the bundle sheath cell, how can the temperature in the mesophyll be lower than that of the bundle sheath cell ? If we stay with the hypothesis that the circulatory system of the leaf is cooling the cells around it, maybe the optimal temperature is one that is between 298°K and 320°K.

→ To investigate further, we need to check how the rate of the two enzymes changes with temperature.

12.5.2 Notes on ETC temperature dependence

The optimal temperature of the Electron Transport Chain (ETC) is thought to be around 25°C or 298°K. However, this has been found to vary depending on the plant's growth temperature [21]. Indeed, depending on the temperature condition during the plant growth, the optimal temperature of the ETC could go as high as 35°C or 308°K in spinach leaves. The paper investigating this phenomenon unfortunately did not test the enzyme rates at higher temperature but this hints at some flexibility when it comes to ETC rate temperature dependence. The paper also states that at high temperatures, the limiting enzyme may be the ETC, and not Rubisco.

→ Maybe the optimal temperature is either one where both the ETC and Rubisco work below their maximum rate but at a rate that is the best compromise for both → Maybe the temperature of the Mesophyll in practice is higher than the optimal temperature for the ETC, which makes the ETC the limiting factor

→ In the case of C4 photosynthesis, the C4 cycle is also relevant for the optimal temperature of the mesophyll. The optimal temperature for PEPC and carbonic anhydrase must be considered.

→ In practice, the enzymes do not work at their theoretical optimal temperature, but often below. Why?? is it a tradeoff of just an artifact??

A paper from 2007 [22] investigates the temperature dependence of the whole photosynthetic chain. At temperatures around 40°C or 313°K, the measured carbon assimilation rate is roughly the same as the one at 25°C or 298°K. However, the optimal temperature for the whole system seems to be lower, around 35°C or 308°K

12.6 Notes on C3 plants with Kranz Anatomy

There are two main cases where C3 plants can exhibit Kranz-like anatomy :

- Proto-Kranz : The Bundle Sheath cells contains more chloroplasts and mitochondria than regular C3 plants, the mesophyll cells are more organised around bundle sheath cells but the plant still exhibit typical C3 biochemistry
- Kranz-like anatomy, with enlarged bundle sheath cells, partial CO₂ concentration mechanisms and reduced photorespiration but still mainly uses the C3 pathway.

Visually and anatomically, the main axis of differentiation is the amount of chloroplasts contained in the Bundle Sheath cells. [23] Going from C3 plant, then Proto-Krantz, Kranz-like and C4, the distribution of chloroplasts between mesophyll and bundle sheath cells is progressively skewed towards a higher amount of chloroplasts in the bundle sheath.

The localization of enzymes like the GDC-P also goes from the Mesophyll to almost exclusively around the vascular bundle. However, looking at the overall amount of relevant enzymes, there is nearly no differences between all C3 plants, including those exhibiting Krantz-like anatomy. There is a very clear separation in enzyme expression profiles between C3 and C4.

12.7 Notes on photosynthesis at low temperatures

At lower temperatures, the C3 pathway presents an advantage over the C4 pathway. As C3 photosynthesis is less energy demanding than C4 under conditions that do not lead to photorespiration. [24] In cold growth conditions, Rubisco specificity is greater for CO_2 than for O_2 [?]

12.8 Anatomical adaptations to extreme temperature

Plants growing in extreme temperature environments have developed different anatomical and physiological adaptations. If we want to model a C4-like C3 plant's anatomical adaptations to extreme temperatures, we need to focus on Bundle Sheath structure, size and placement, as well as the number of mesophyll cells between bundle sheath cells and the surface of the cell. Some C3 plants already exhibit Rubisco activity in Bundle sheath cells. The main difference between other models and our C4 model would then be the density, or number of Bundle Sheath cells and Meso,phyll cells. Usually in C4 plants, there is only one layer of both. In C3 plants, this is more variable. This addition of cell layers could be translated in the model by less energy transfers (lower UA).

12.9 Ideas on outliers

→ Some C4 plants have 2 layers of bundle sheath cells; some use the outer for the Calvin cycle, and some the inner. Why ? → C3 plants work better in low-temperature environments because they spend less energy without having to concentrate carbon. The lower temperature drastically reduces the photorespiration rate.

12.10 Tsoil, Tsky, Solar irradiance, Tair

The mean solar irradiance on top of the atmosphere is $1361 W.m^{-2}$

The mean temperature of the soil in Soudan goes from 20°C to 40°C

==> Need to find the relationship between irradiance and soil, air and sky temperature $T_{air}, T_{soil}, T_{sky} = f(\phi)$

Equations for relationships of T_{air} and T_{soil} with ϕ from empirical data : 01.08.2013 in Sudan.

$$T_{air}(\phi) = 0.077\phi + 28 \quad (12.8)$$

Soil Temperature variation depends on both soil water content (humidity) and irradiance.

12.11 Notes on Rubisco dependance on Rubisco activase

Two experiments were performed to asses the role of Rubisco activase on its catalytic efficiency. They measured CO₂ assimilation rate in C₄ plant *Flaveria Bidentis* inhibited expresion of rubisco activase. The results were consistent between the two papers [25] [26]. They find that when Rubisco activase expression is highly inhibited, the CO₂ assimilation rate is strongly reduced. This effect is shown at both atmospheric CO₂ pressures and at high (up to 2.5kPa pCO₂) CO₂ pressures. However, at high temperatures of around 40°C, the activity of the Rubisco activase was strongly reduced regardless of external inhibition, without affecting CO₂ assimilation rate. This indicates that other factors are at play. Authors point to the impact of Mg²⁺ as a limiting factor at high temperatures and high CO₂ pressures. [26]. Overall, the two papers point to a reduced Rubisco activity if Rubisco activase concentration reaches below 25% of Wild Type concentration, with little to no effect above that. Given that the Rubisco activase is irreversibly denatured (destroyed) beyond 45°C, this should be considered a high bound for optimal temperature of Rubisco. [27].

12.12 Notes on PSII temperature dependance

The temperature dependance of Photosystem II is apparently more difficult to measure than that of other enzymes. Some measures have still been done and it seems that the complex works best around 30°C. [28]

12.13 Notes on PEPC temperature dependance

PEPC temperature dependance changes a lot depending on wether or not it's expressed by a C₃ plant or a C₄ plant. PEPC in C₄ plants has a much higher tolerance to high temperatures, as well as a higher optimal temperature. Also, C₄ PEPC is much more susceptible to denaturation at low temperatures than C₃ PEPC. The optimal temperature for the enzyme would be between 35 and 50°C (Optimal temperature at 45°C) [29]

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Table 5: Summary of Variables and Parameters in the C4 Photosynthesis Model

Symbol	Description	Units	Value (if applicable)
A	CO ₂ assimilation rate	$\mu\text{mol m}^{-2} \text{s}^{-1}$	-
V_c	Rate of Rubisco carboxylation	$\mu\text{mol m}^{-2} \text{s}^{-1}$	-
V_o	Rate of Rubisco oxygenation	$\mu\text{mol m}^{-2} \text{s}^{-1}$	-
R_d	Mitochondrial respiration rate	$\mu\text{mol m}^{-2} \text{s}^{-1}$	-
$R_{d,m}$	Mesophyll mitochondrial respiration rate	$\mu\text{mol m}^{-2} \text{s}^{-1}$	-
$R_{d,bs}$	Bundle sheath mitochondrial respiration rate	$\mu\text{mol m}^{-2} \text{s}^{-1}$	-
V_p	Rate of PEP carboxylation	$\mu\text{mol m}^{-2} \text{s}^{-1}$	-
R_m	Mesophyll mitochondrial respiration	$\mu\text{mol m}^{-2} \text{s}^{-1}$	-
L	CO ₂ leakage rate	$\mu\text{mol m}^{-2} \text{s}^{-1}$	-
g_{bs}	Conductance to CO ₂ leakage	$\text{mol m}^{-2} \text{s}^{-1} \text{Pa}^{-1}$	0.003
C_s	Bundle sheath CO ₂ partial pressure	μbar	-
C_m	Mesophyll CO ₂ partial pressure	μbar	-
O_s	Bundle sheath O ₂ partial pressure	mbar	-
O_m	Mesophyll O ₂ partial pressure	mbar	-
Γ^*	CO ₂ compensation point	μbar	0.0003817
$S_{c/o}$	Rubisco specificity	-	-
$V_{c\max}$	Maximum Rubisco activity	$\mu\text{mol m}^{-2} \text{s}^{-1}$	40
K_c	Michaelis constant of Rubisco for CO ₂	μbar	1210
K_o	Michaelis constant of Rubisco for O ₂	mbar	292
$V_{p\max}$	Maximum PEPC activity	$\mu\text{mol m}^{-2} \text{s}^{-1}$	200
K_p	Michaelis constant of PEPC for CO ₂	μbar	82
γ	Energy requirement due to photorespiration	-	-
J	Rate of electron transport	$\mu\text{mol m}^{-2} \text{s}^{-1}$	-
$J_{\max}$	Maximum electron transport rate ⁴³	$\mu\text{mol m}^{-2} \text{s}^{-1}$	248
θ	Curvature factor for electron transport	-	0.7
I	Photosynthetically useful light absorbed by PSII	$\mu\text{mol m}^{-2} \text{s}^{-1}$	-